FOREIGN TRADE UNIVERSITY

FINANCE & BANKING FACULTY

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MIDTERM REPORT

RELATIONSHIP BETWEEN CORPORATE GOVERNANCE AND FIRM PROFITABILITY

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ABSTRACT

This study examines the impact of corporate governance on the financial performance (Pretax ROA and ROE) of 21 firms listed on the Hanoi Stock Exchange (HNX) from 2010 to 2024. Focused on the Pharmaceutical and Real Estate sectors, the research utilizes a Fixed Effects (FE) model with clustered robust standard errors.

Findings indicate that while governance significantly influences profitability, effects are highly sector-dependent. For the full sample, board diligence (meeting frequency), CEO age, and Big 4 auditors are robust positive drivers, whereas larger board sizes diminish efficiency.

Sectoral analysis reveals that governance is more influential in Pharmaceuticals, where CEO duality and leverage negatively impact performance, while female leadership and gender diversity are positive predictors. Conversely, Real Estate profitability is driven more by financing cycles and leverage than internal governance. The results suggest that regulatory frameworks should be sector-specific rather than "one-size-fits-all."

Keywords: Corporate Governance, Firm Performance, Pharmaceutical Industry, Real Estate Industry, Vietnam, Fixed Effects Model.

INTRODUCTION

1. Rationale of the study

Corporate governance has become an important determinant of firm performance, particularly in emerging markets where institutional frameworks are still developing. In Vietnam, regulatory reforms have strengthened governance standards for listed firms through governance codes and improved disclosure (World Bank, 2019; IFC, 2018). However, empirical evidence on the governance-profitability relationship remains mixed and context-dependent (Nguyen, Locke and Reddy, 2015; Tran and Pham, 2020).

Many prior studies in Vietnam use pooled industry samples without considering sectoral differences, although industry characteristics such as capital intensity, leverage, and regulatory pressure may influence governance effectiveness (Klapper and Love, 2004; Hillman, Withers and Collins, 2009). In this context, the pharmaceutical and real estate sectors represent structurally different industries suitable for comparative analysis. Therefore, this study examines the relationship between corporate governance and profitability using a sample of 21 firms listed on the Hanoi Stock Exchange (HNX) during the period 2010-2024.

2. Research Objectives

To achieve the general objective, the study pursues the following specific objectives:

- To assess how board structure characteristics (board size, independence, gender diversity, leadership duality, and meeting frequency) influence firm profitability.
- To evaluate the impact of executive characteristics (chairperson and CEO gender and age) on firm performance.
- To analyze whether governance quality proxies, such as foreign ownership and the engagement of Big 4 auditors, affect profitability.

- To compare the effects of governance variables on two profitability indicators: pre-tax return on assets (pretax ROA) and pre-tax return on equity (pretax ROE).
- To examine whether the relationship between corporate governance mechanisms and firm profitability differs across industries, specifically between the real estate sector and the medical sector.

3. Object and Scope of the Study

3.1 Research Object

The research object of this study is corporate governance mechanisms and firm profitability in companies listed on the Hanoi Stock Exchange (HNX), specifically within the pharmaceutical and real estate industries. The study focuses on internal governance structures at the board and executive levels, together with selected external governance factors, and examines their relationship with accounting-based measures of firm performance.

3.2 Research Scope

The scope of this study is defined across four dimensions. First, the subject scope focuses on firms in the pharmaceutical and real estate industries. Second, the market scope includes 21 companies listed on the Hanoi Stock Exchange (HNX). Third, the time scope covers panel data from 2010 to 2024, enabling longitudinal analysis. Finally, the analytical scope examines accounting-based performance measures and internal corporate governance mechanisms, while controlling for firm characteristics such as size, leverage, revenue, and asset structure.

4. Research Methodology

This study employs a **quantitative research design** to investigate the relationship between corporate governance mechanisms and firm profitability among listed firms on the Hanoi Stock Exchange (HNX). **Panel data analysis** is employed to exploit both cross-sectional and time-series variations, allowing

for a more comprehensive assessment of governance-performance dynamics over time.

5. Research Structure

This study is organized into five chapters.

- Chapter 1: Theoretical Framework and Literature Review.
- Chapter 2: Research Methodology
- Chapter 3: Research Results
- Chapter 4: Discussion
- Chapter 5: Recommendations

CHAPTER 1. THEORETICAL FRAMEWORK & LITERATURE REVIEW

1.1. Theoretical Framework

1.1.1. Research Background

1.1.1.1. Importance of Corporate Governance

Corporate Governance (CG) is a foundational element in shaping the operational efficiency and sustainability of listed companies, particularly within dynamic markets. For firms listed on the Hanoi Stock Exchange (HNX), a robust CG structure is crucial for mitigating information asymmetry and agency conflicts. These issues are especially pertinent given Vietnam's evolving legal framework, concentrated ownership structures, and limited minority shareholder protection.

1.1.1.2. Rationale for Comparing the Pharmaceutical and Real Estate Industries

This study compares the pharmaceutical and real estate sectors because industry characteristics fundamentally influence governance structures and performance outcomes. The pharmaceutical industry is highly research-intensive, requiring significant investment in research and development (R&D) and long product development cycles due to complex clinical trials and regulatory approval processes (EFPIA, 2025; ICH, 2023). Although revenues can become stable once products are commercialized, firms must continuously invest in innovation and comply with strict regulatory standards, which increases the importance of strategic oversight and expertise at the board level (Umbrex, 2023).

In contrast, the real estate industry is highly capital-intensive, characterized by substantial debt usage, long-term development projects, and strong sensitivity to macroeconomic conditions and interest rate fluctuations. Consequently, financial risk management and monitoring mechanisms play a crucial role in this sector.

These structural differences suggest that governance needs vary across industries. Real estate firms may require stronger monitoring mechanisms to manage leverage and investment risk, whereas pharmaceutical firms may benefit more from diverse and knowledgeable boards that support innovation and regulatory compliance. Comparing these two sectors therefore allows the study to examine whether the relationship between corporate governance and firm profitability is industry-specific.

1.1.2. Theoretical foundation

The relationship between corporate governance and firm profitability is explained through multiple theoretical perspectives. This study primarily adopts Agency Theory while also drawing on Resource Dependence Theory to understand how governance mechanisms influence firm performance across industries.

1.1.2.1. Agency Theory

Agency Theory, introduced by Michael C. Jensen and William H. Meckling (1976), provides the primary framework for examining the relationship between corporate governance and firm performance. The theory argues that the separation of ownership and control in modern corporations creates conflicts of interest between shareholders (principals) and managers (agents), generating agency costs such as monitoring, bonding, and residual losses. If not properly controlled, these conflicts can reduce firm value and profitability. In publicly listed firms, dispersed ownership and professional management intensify agency problems, as managers may pursue opportunistic behaviors such as earnings manipulation, overinvestment, inefficient capital allocation, or excessive risk-taking. Corporate governance mechanisms are therefore designed to mitigate these problems by strengthening monitoring systems and aligning managerial incentives with shareholder interests.

Board characteristics represent important internal governance mechanisms. Board independence improves objective oversight, while appropriate board

size and frequent meetings enhance monitoring effectiveness. Separating the CEO and chair roles reduces power concentration and increases accountability, whereas CEO duality may weaken board oversight and negatively affect firm performance.

1.1.2.2. Resource Dependence Theory

Resource Dependence Theory (Pfeffer & Salancik, 1978) views the board of directors not only as a monitoring body but also as a strategic mechanism that helps firms manage external dependencies and access critical resources such as capital, information, and legitimacy. Board members contribute expertise, reputation, and networks that can enhance strategic decision-making and firm performance. In this study, foreign ownership and audit quality reflect these external linkages, as foreign investors may provide international capital and managerial practices, while reputable audit firms enhance credibility and reduce information asymmetry.

This perspective also highlights industry differences. In pharmaceutical firms, characterized by high R&D intensity and strict regulation, boards with technical expertise can support innovation and compliance. In contrast, real estate firms rely heavily on external financing, making monitoring mechanisms particularly important. Consequently, the governance-profitability relationship may vary across industries.

1.2. Definitions

1.2.1. Independent variables: Corporate governance

Corporate governance is defined as the system by which companies are directed and controlled, specifying the distribution of rights and responsibilities among different participants in the corporation, including the board of directors, managers, shareholders, and other stakeholders, and providing the structure for corporations to set objectives and monitor performance (Organisation for Economic Co-operation and Development, 2015).

From an agency theory perspective, corporate governance mechanisms serve to mitigate conflicts of interest between managers (agents) and shareholders (principals), thereby reducing agency costs and improving firm performance (Jensen & Meckling, 1976; Fama & Jensen, 1983).

Based on these theoretical foundations, this study operationalizes corporate governance through board structure and leadership characteristics.

1.2.1.1. Board structure

Board Size: refers to the total number of people serving on the board. According to resource dependence theory, larger boards may provide diverse expertise, broader external connections, and improved access to resources (Pfeffer & Salancik, 1978). However, agency theory suggests that excessively large boards may reduce coordination efficiency and monitoring effectiveness (Yermack, 1996). Therefore, board size reflects a trade-off between advisory capacity and decision-making efficiency.

Board Independence: defined as the presence or proportion of non-executive independent directors on the board. Independent directors are those who do not participate in daily management and have no material relationship with the firm. Fama and Jensen (1983) argue that independent directors strengthen monitoring functions and reduce managerial opportunism. The OECD (2015) also emphasizes the importance of independent oversight in ensuring objective judgment in corporate decision-making.

Female Board Representation: captures gender diversity at the board level. Board diversity is associated with improved monitoring quality, ethical awareness, and decision-making processes (Adams & Ferreira, 2009). From a governance perspective, diversity enhances board independence and reduces groupthink, thereby potentially improving firm performance.

CEO Duality: occurs when the Chief Executive Officer simultaneously holds the position of Chairperson of the board. Agency theory argues that separating the roles of CEO and Chair enhances monitoring effectiveness by preventing

concentration of decision-making power (Jensen, 1993). In contrast, stewardship theory suggests that unified leadership may enhance strategic consistency and efficiency. CEO duality, therefore, reflects the balance between managerial power and board oversight.

Number of Board Meetings: represents the frequency of board activity within a fiscal year. Vafeas (1999) suggests that more frequent meetings signal active monitoring and stronger governance engagement. The OECD (2015) similarly highlights the importance of board diligence in fulfilling oversight responsibilities.

1.2.1.2. Executive characteristics

Chair Gender and CEO Gender: reflecting leadership diversity. Upper echelons theory posits that organizational outcomes partially reflect the characteristics of top executives (Hambrick & Mason, 1984). Gender diversity may influence risk preferences, ethical standards, and strategic decisions, thereby affecting firm performance (Post & Byron, 2015).

Chair Birth Year and CEO Birth Year (Executive Age): captures managerial experience and risk tolerance. Younger executives tend to exhibit greater risk-taking behavior, while older executives may adopt more conservative strategies (Serfling, 2014). Under the upper echelons theory, demographic characteristics influence corporate policies and performance outcomes.

1.2.2. Dependent variables: Firm profitability

Firm profitability refers to a firm's ability to generate earnings from its resources and equity base. It reflects operational efficiency and value creation for shareholders. This study employs:

Pre-tax Return on Assets (Pretax ROA): defined as earnings before tax divided by average total assets of two consecutive years. It measures managerial efficiency in utilizing firm assets to generate operating profits.

ROA is widely used in corporate governance research because it reflects internal performance independent of financing structure (Zeitun & Tian, 2007).

Pre-tax Return on Equity (Pretax ROE): defined as earnings before tax divided by average shareholders' equity of two consecutive years. It measures the return generated for equity holders and reflects the firm's effectiveness in creating shareholder value.

The use of pre-tax profitability measures eliminates distortions arising from variations in tax policies and tax management strategies (Hanlon & Heitzman, 2010). Employing both ROA and ROE ensures robustness, as ROA captures operational efficiency while ROE incorporates financial leverage effects.

1.2.3. Control variables

Control variables are incorporated to account for firm-specific characteristics that influence profitability and may confound the relationship between governance and performance.

Firm Size: measured as the natural logarithm of total assets. Larger firms typically benefit from economies of scale, greater diversification, and stronger access to capital markets (Titman & Wessels, 1988). Logarithmic transformation reduces skewness and mitigates the influence of extreme values.

Leverage: measured as total debt divided by total assets (TD/TA) or total liabilities divided by total assets (TL/TA). According to capital structure theory (Modigliani & Miller, 1963), leverage influences firm performance through tax shields and financial distress risk.

Foreign Ownership: reflects the participation of foreign investors in the firm. Foreign shareholders are often associated with higher governance standards, improved transparency, and stronger monitoring mechanisms (Choi, Sul & Min, 2012).

Big4 Auditing: indicates whether the firm is audited by one of the major international auditing firms, including Deloitte, KPMG, Ernst & Young, or PricewaterhouseCoopers. Engagement of Big4 auditors is associated with

higher audit quality, reduced information asymmetry, and improved financial reporting credibility (Francis & Yu, 2009).

Sales Growth: measures the annual percentage change in revenue and captures firm expansion and market performance (Evans, 1987). This research calculates sales growth by dividing the difference in sales between two consecutive years by the former year's sales.

Asset Tangibility: defined as gross property, plant, and equipment divided by total assets. Tangible assets influence collateral value, borrowing capacity, and operational structure (Rajan & Zingales, 1995).

1.3. Literature Review

A complicated interaction between internal oversight and market-specific factors is revealed by the empirical landscape surrounding the effect of corporate governance on business profitability. In order to create a framework for examining HNX-listed companies in the real estate and pharmaceutical industries, this section summarizes the results of important studies.

1.3.1. Core Governance Determinants and Profitability in Vietnam

Empirical evidence from Vietnam's transition economy shows that board-level practices play an important role in reducing agency conflicts and improving financial performance. Studies on firms listed on the Hanoi Stock Exchange (HNX) indicate that board size and board independence significantly influence ROE and ROA. Consistent with Resource Dependence Theory, prior research suggests that larger and more diverse boards can enhance resource access and strategic guidance (Anh, H.D.V., Huy, D.N.M., & Phuong, N.H.B., 2020). However, since this relationship is often non-linear, maintaining an optimal board size is necessary to prevent communication inefficiencies that may reduce profitability (Vo, D.H. & Nguyen, T.M., 2014).

Moreover, the impact of CEO duality remains empirically debated. Evidence in the Vietnamese context suggests that combining the CEO and chairman roles often leads to managerial entrenchment, weakening board oversight and

negatively affecting firm performance (Le, H.T.M., Ting, I.W.K., Kweh, Q.L., & Ngo, H.L.T., 2023). In contrast, board meeting frequency appears to improve outcomes, as firms that meet more often tend to be more profitable due to stronger supervision and active monitoring (Le, T.M.H., Ting, I.W.K., Kweh, Q.L., & Hoanh, L.T.H., 2018).

1.3.2. The Interplay between Governance, Leverage, and Control Variables

A substantial body of research examines how governance quality moderates the relationship between financial structure and profitability. In Vietnam, strong governance practices are shown to act as a "buffer," mitigating the negative effects of excessive financial leverage on firm performance (Pham, H.S.T., & Nguyen, D.T., 2020). This suggests that firms with effective governance are better able to manage debt and interest obligations without negatively affecting ROA.

There is strong evidence supporting the inclusion of control variables such as firm size, foreign ownership, and audit quality in empirical models. Larger firms often benefit from economies of scale, leading to more stable profitability. Moreover, the presence of Big 4 audit firms and foreign institutional investors signals greater transparency and oversight. These external monitors are associated with higher ROE because they promote global best practices and reduce agency costs commonly faced by firms in emerging markets (Anh, H.D.V., Huy, D.N.M., & Phuong, N.H.B., 2020).

1.3.3. Sectoral Dynamics: A Comparative Empirical Perspective

The effectiveness of corporate governance is strongly influenced by industry-specific risk profiles and capital structures. This study highlights a clear distinction between the pharmaceutical and real estate sectors.

Empirical evidence suggests that financial performance in the real estate sector largely depends on managing agency costs associated with high capital intensity and significant debt levels. Governance mechanisms therefore need

to prioritize oversight of debt-related risks and asset substitution (Nguyen, A.T., Tran, D.T.T., & Vuong, A.T., 2026). As a result, variables such as Leverage and Firm Size often show greater statistical significance in predicting ROA than internal board characteristics.

Conversely, the pharmaceutical industry requires a governance model focused on strategic stability and risk reduction. Because pharmaceutical firms rely on long-term operating cycles and intellectual assets, their profitability is highly sensitive to the quality of internal governance (Liu, W. & Li, R., 2024). Unlike real estate - where performance often follows external credit cycles - ROA in the pharmaceutical sector is more closely linked to the board's ability to maintain operational transparency and strategic continuity. These findings support the use of sub-sample regressions and sectoral dummy variables to capture distinct industry effects in the dataset.

1.3.4. Conclusion of Empirical Studies

Although its impact varies by governance structure and industry context, prior empirical evidence generally shows that corporate governance plays a key role in determining firm profitability in Vietnam. While financial leverage and firm-specific control variables also influence this relationship, board characteristics, CEO duality, and monitoring intensity consistently emerge as major predictors of ROA and ROE. Sectoral differences between pharmaceutical and real estate firms further highlight the importance of industry-specific risk profiles and capital structures in explaining performance. These mixed and context-dependent findings therefore justify further empirical research focusing on HNX-listed firms in the pharmaceutical and real estate sectors.

CHAPTER 2. RESEARCH METHODOLOGY

2.1. Hypothesis Development and Research Model

2.1.1. Hypothesis Development

In order to explain how governance systems affect business profitability, this study's conceptual framework combines Agency Theory and Resource Dependence Theory. While Resource Dependence Theory (Pfeffer & Salancik, 1978) suggests that a larger board provides more strategic resources, evidence in the Vietnamese context often shows a non-linear relationship, where excessively large boards reduce coordination efficiency. Therefore, this study proposes that board size beyond an optimal level may negatively affect profitability. In contrast, Board Independence is expected to act as an important monitoring mechanism that reduces agency costs and improves ROE and ROA. Moreover, board meeting frequency is assumed to enhance performance by reflecting active participation and closer supervision of management.

Based on these theoretical foundations, the following hypotheses are proposed:

H1: Board size has a significant impact on firm profitability, with potential negative effects if the board becomes too large to coordinate effectively.

H2: Board independence is positively associated with firm profitability (ROA and ROE) due to enhanced monitoring.

H3: The frequency of board meetings has a positive impact on firm performance.

H4: Gender diversity within the board (female board members) positively influences firm profitability.

The study also emphasizes the role of external monitoring and executive characteristics. Based on Agency Theory (Jensen & Meckling, 1976), CEO Duality may reduce firm performance by concentrating power and weakening the board's oversight. CEO Age (current year − birth year) is expected to be a key demographic predictor, as older and more experienced CEOs are assumed to better manage market complexities and improve returns. In addition, external

governance proxies—foreign ownership and Big 4 auditors—are expected to positively relate to profitability because they enhance transparency and enforce international best practices, which are important for firms listed on the Hanoi Stock Exchange (HNX). Finally, the relationship between these governance factors and profitability is expected to vary by industry, reflecting structural differences between the pharmaceutical and real estate sectors.

Accordingly, we hypothesize:

H5: CEO Duality is negatively correlated with firm profitability.

H6: CEO age (measured by current year - birth year) is a significant predictor of performance, with older CEOs being more effective in optimizing returns.

H7: Foreign ownership and the involvement of Big 4 auditors are positively correlated with firm profitability.

H8: The relationship between corporate governance mechanisms and firm profitability differs significantly between the pharmaceutical and real estate industries.

2.1.2. Research Model

Panel data analysis is used in this study's quantitative design to evaluate these hypotheses empirically. This method enables the analysis of governance-performance dynamics over a number of time periods for 21 companies listed on the HNX. The most suitable analytical tool was found to be the Fixed Effects Model (FEM) with Cluster-Robust Standard Errors, according to diagnostic testing, which included the Modified Wald test, which showed the presence of heteroskedasticity. The results could be skewed by unobserved firm-specific heterogeneities, which our model successfully accounts for.

The primary regression model is formulated as follows:

$$Profitability_{it} = \beta_0 + \beta_1 CG_{it} + \beta_2 Controls_{it} + \gamma_t Year_t + u_i + \varepsilon_{it}$$

In this specification, $Profitability_{it}$ represents the dependent variables, pre-tax ROA and pre-tax ROE. The vector CG_{it} encompasses the core independent variables: boardsize, independentboardmember, femaleboardmember, duality,

numberofboardmeeting, chair_gender, chair_age, ceo_gender, ceo_age, foreign_ownership, and big4. To ensure the robustness of the findings, the model incorporates several firm-level control variables ($Controls_{it}$), specifically Firm Size (*ln_size*), Financial Leverage (*leverage*), Sales Growth, and Fixed Asset Ratio. These variables are essential to isolate the specific impact of corporate governance from other economic drivers of profitability.

2.2. Data Collection

This study employs a balanced and unbalanced panel dataset comprising 21 firms listed on the Hanoi Stock Exchange (HNX), operating in two distinct industries: pharmaceutical and real estate. The sample was determined based on the availability of consistent financial and corporate governance information over the study period.

Financial data, including total assets, total liabilities, total debt, total revenues, gross property, plant and equipment (PPE), and profitability indicators (pre-tax ROA and pre-tax ROE), were collected and calculated from audited annual financial statements published by the firms. Corporate governance variables - such as board size, board independence, gender composition, CEO duality, number of board meetings, and executive demographic characteristics - were manually compiled from annual reports and corporate disclosures.

After data cleaning and transformation procedures, including the treatment of missing values and outliers, the final dataset consists of 248 firm-year observations. The panel structure allows for the examination of both cross-sectional and time-series variations, thereby enhancing the reliability of causal inference.

To mitigate the influence of extreme observations, key financial variables (pretax ROA, pretax ROE, and leverage) were winsorized at the 1st and 99th percentiles. Firm size was transformed using the natural logarithm of total assets to reduce skewness and ensure normality. Panel structure was established using firm identifiers and year variables, enabling fixed-effects estimation.

2.3. Variables Measurement

This study categorizes variables into three groups: dependent variables (firm profitability), independent variables (corporate governance mechanisms), and control variables (firm-specific characteristics).

2.3.1. Dependent Variables

Pre-tax Return on Assets (Pretax ROA): Earnings before tax divided by average total assets of two consecutive years. This measure captures managerial efficiency in utilizing firm assets to generate operating income.

Pre-tax Return on Equity (Pretax ROE): Earnings before tax divided by average shareholders' equity. This metric reflects the return generated for equity holders and incorporates the impact of financial leverage.

Using pre-tax indicators avoids distortions caused by tax policy differences and tax management strategies, ensuring greater comparability across firms.

2.3.2. Independent Variables: Corporate Governance

Corporate governance is proxied through board structure and executive characteristics:

- **Board Size (boardsize):** Total number of board members.
- **Board Independence (independentboardmember):** Number of independent directors.
- **Female Board Representation (femaleboardmember):** Number of female board members.
- **CEO Duality (duality):** Dummy variable equal to 1 if the CEO simultaneously serves as board chair.
- **Number of Board Meetings (numberofboardmeeting):** Annual meeting frequency.
- **Chair Gender (chair_gender):** Dummy variable representing chairperson's gender (male = 1, female = 0)
- **Chair Age (chair_age):** Proxy for chair age.

- **CEO Gender (ceo_gender):** Dummy variable representing CEO gender (male = 1, female = 0)
- **CEO Age (ceo_age):** Proxy for CEO age.

2.3.3. Control Variables

To isolate the effect of governance mechanisms, the following firm-specific control variables are incorporated:

- **Foreign Ownership (foreign_ownership):** Dummy variable indicating foreign investor participation (1 if the firm has foreign ownership, 0 otherwise)
- **Big4 Auditing (big4):** Dummy variable equal to 1 if the firm is audited by Deloitte, KPMG, EY, or PwC.
- **Firm Size (ln_size):** Natural logarithm of total assets.
- **Leverage (leverage):** Total liabilities divided by total assets.
- **Sales Growth (sales_growth):** Annual percentage change in total revenues.
- **Fixed Asset Ratio (fixed_asset_ratio):** Gross PPE divided by total assets.

2.4. Analytical Methods

2.4.1. Panel Data Estimation

Given the longitudinal structure of the dataset, panel data regression techniques are employed. Three primary models were considered:

- **Pooled Ordinary Least Squares (OLS)**
- **Random Effects Model (REM)**
- **Fixed Effects Model (FEM)**

Model selection was conducted using the Hausman test and additional diagnostic procedures.

The general empirical specification is:

$$Profitability_{it} = \beta_0 + \beta_1 CG_{it} + \beta_2 Controls_{it} + \gamma_t Year_t + \mu_i + \varepsilon_{it}$$

Where:

i denotes firm

t denotes year

μ_i captures unobserved firm-specific effects

γ_t controls for time effects

2.4.2. Diagnostic Tests

Several diagnostic tests were performed to ensure model validity:

- **Multicollinearity Test:** Variance Inflation Factor (VIF)
- **Heteroskedasticity Test:** Modified Wald test (xttest3)
- **Autocorrelation Test:** Wooldridge test (xtserial)
- **Model Selection:** Hausman test and F-test for fixed effects

Results indicate the presence of heteroskedasticity but no first-order autocorrelation. Therefore, all final estimations were conducted using the Fixed Effects Model with cluster-robust standard errors at the firm level (vce(cluster firm)).

2.4.3. Sub-Sample and Comparative Analysis

To examine industry-specific effects, the sample was divided into two sub-groups:

- Pharmaceutical firms (BDS = 0)
- Real estate firms (BDS = 1)

Separate fixed-effects regressions were estimated for each sub-sample to identify structural differences in governance-performance relationships.

CHAPTER 3. RESEARCH RESULTS

3.1. Descriptive Statistics and Correlation Matrix

3.1.1 Descriptive Statistics

Table 1. Descriptive Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
pretax_roe	248	0.1678811	0.1803338	-0.2923875	0.65
pretax_roa	248	0.0917726	0.1052861	-0.1961	0.3937
boardsize	247	5.574899	1.200211	3	12
independentboard member	220	1.731818	1.422547	0	8
femaleboardmemb er	246	1.150407	0.8557965	0	3
duality	247	0.1983806	0.39959	0	1
numberofboardme eting	194	10.21649	6.999225	1	43
chair_gender	243	0.872428	0.3343012	0	1
chair_age	238	53.2521	9.384176	32	73
ceo_gender	241	0.846473	0.361245	0	1
ceo_age	233	50.06438	9.011495	31	71
ln_size	248	15.76335	5.12459	5.645366	27.27132
leverage	248	0.4326969	0.20696	0.0529105	0.8384427
sales_growth	221	0.3381874	1.407745	-0.9761704	8.613344

foreign_ownership	239	0.8033473	0.3983017	0	1
big4_auditing	246	.0853659	0.279995	0	1
fixed_asset_ratio (PPE)	248	0.1982678	0.2408065	0.00000838	1.038315

Source: Aggregated result from Stata

This study uses an unbalanced panel dataset of 21 firms listed on the HNX observed over 2010-2024. The dataset covers firm profitability measures, corporate governance characteristics, and a set of control variables commonly used in the corporate governance literature.

Regarding firm performance, the mean Pretax ROE is 0.168 with a standard deviation of 0.180, indicating substantial dispersion in returns to shareholders across firms and years. The minimum value of -0.292 implies that several firms experienced meaningful losses in certain years, while the maximum value of 0.650 reflects strong profitability during peak periods. Similarly, mean Pretax ROA equals 0.092 (SD = 0.105), suggesting that firms' operating profitability also varies considerably over time. Notably, ROE exhibits greater variability than ROA, which is consistent with ROE's higher sensitivity to leverage and changes in the equity base.

In terms of governance characteristics, the average board size is 5.57 members (range: 3-12), which is consistent with typical board structures among Vietnamese listed companies. On average, firms have 1.73 independent directors and 1.15 female directors, suggesting modest independence and gender diversity in the sample. CEO duality is observed in approximately 19.8% of firm-year observations, indicating that most firms separate the CEO and chair roles. Boards hold 10.22 meetings per year on average, but the range (1-43) indicates strong heterogeneity in monitoring intensity. Executive

demographics suggest that leadership is generally mature: mean chair age is 53.25 years and mean CEO age is 50.06 years.

With respect to control variables, the mean leverage ratio is 0.433, implying that liabilities finance roughly 43% of total assets on average. Firm size (log total assets) shows substantial variation across the sample, reflecting heterogeneous scale. Sales growth is volatile (mean = 0.338; SD = 1.408), indicating that revenue expansion is uneven across firms and years. Foreign ownership is relatively prevalent (mean = 0.803), whereas Big4 auditing is comparatively rare (mean = 0.085). Finally, the fixed asset ratio (PPE/TA) displays wide dispersion, which is consistent with structural differences between more asset-intensive real estate firms and less tangible-asset-dependent pharmaceutical firms. Overall, the descriptive statistics indicate meaningful cross-sectional and time variation in key variables, supporting the use of panel regression techniques in subsequent analysis.

3.1.2 Correlation Matrix

Table 2. Correlation Matrix

Variable	Pretax roa	Pretax roe	Board size	Independent board member	Female board member	Duality	Number of board meeting
Pretax roa	1						
Pretax roe	0.8267***	1					
Board size	0.2628***	0.2250***	1				
Independent board member	-0.1673**	-0.1374**	0.0213	1			
Female board	0.0590	-0.0118	0.2406***	0.1060	1		

member							
duality	0.0490	0.0506	-0.0692	-0.1457**	0.0909	1	
Number of board meeting	-0.0487	-0.0203	-0.1082	-0.0967	-0.0596	-0.1514**	1
Chair gender	0.0577	0.0641	-0.1111*	-0.1419**	-0.302***	-0.0230	0.0610
Chair age	0.1895***	0.1678***	0.2525***	-0.1392**	0.1807***	-0.0135	-0.0061
Ceo gender	-0.0308	-0.0016	0.0362	-0.1236*	-0.296***	0.0150	-0.1084
Ceo age	0.3988***	0.3354***	0.2580***	-0.1785**	0.1384**	0.1663**	-0.0812
ln_size	-0.181***	-0.1629**	-0.1297**	0.5928***	0.1937***	-0.0805	-0.1205*
leverage	-0.1501**	0.1869***	0.1487**	0.1316*	-0.0233	0.0094	-0.0114
Sales growth	0.0263	0.0551	0.0074	-0.0565	-0.0327	0.1638**	-0.0033
Foreign ownership	-0.1422**	-0.1939** *	-0.2340** *	0.1149*	-0.1397**	-0.1198*	-0.0601
Big4 auditing	-0.177***	-0.185***	-0.1342**	0.3306***	-0.0708	-0.1152*	-0.1146
(PPE)	0.0901	0.0096	-0.0647	0.2956***	0.0875	0.1866***	-0.289***

**** p<0.01, ** p<0.05, * p<0.1*

Source: Aggregated result from Stata

Table 2 presents the pairwise correlations among the main profitability measures and corporate governance variables, while the full correlation matrix including control variables is reported in Appendix 1.

First, both ROE and ROA show statistically significant associations with several firm fundamentals, including firm size (ln_size) and leverage, suggesting that firm scale and capital structure are systematically related to performance in the bivariate setting.

Second, within corporate governance variables, several significant correlations indicate that governance mechanisms tend to co-move rather than operate independently. For example, board size is significantly associated with board composition and leadership characteristics (including female representation and executive age proxies), while board independence is significantly related to leadership structure, most notably showing a negative association with CEO duality. This pattern is consistent with the idea that stronger monitoring structures are less likely to coexist with concentrated leadership arrangements.

Third, governance variables also correlate significantly with external monitoring and firm characteristics. In particular, firm size (ln_size) is strongly positively correlated with board independence, and it is also significantly associated with external monitoring proxies such as Big4 auditing and asset structure (fixed_asset_ratio). This suggests that larger firms in the sample tend to exhibit stronger monitoring environments and different asset compositions, which is consistent with scale-related differences in governance complexity and disclosure practices.

Finally, although several statistically significant correlations are observed, most coefficients remain moderate in magnitude, reducing concerns about severe pairwise multicollinearity. This is consistent with the multicollinearity diagnostics reported later (Mean VIF = 4.77).

3.2. Diagnostic Tests and Model Selection

3.2.1. Multicollinearity Test

Purpose: To examine whether the independent variables in the model are highly correlated, which could distort the regression coefficients.

Method: Variance Inflation Factor (VIF).

Decision Rule: If VIF < 10 (or strictly < 5), there is no severe multicollinearity.

Result: Mean VIF = 4.77. The highest VIF for a non-year variable is 4.08 (ln_size).

Conclusion: All values are well below the threshold of 10. Multicollinearity is not a significant concern; the null hypothesis is accepted.

3.2.2. Heteroskedasticity Test

Purpose: To test whether the variance of the error terms is constant across observations.

Hypotheses:

H0: Var(error) is constant across all observations (Homoskedasticity).

H1: Var(error) is not constant across all observations (Heteroskedasticity).

Method: Modified Wald test for groupwise heteroskedasticity in Fixed Effect regression model (Command: xttest3).

Results: Prob > chi^2 = 0.0000

Conclusion: The p-value is significantly lower than 0.01; thus, the null hypothesis is rejected at the 1% significance level. Heteroskedasticity is present in the panel data.

3.2.3. Autocorrelation Test

Purpose: To check for the presence of first-order serial correlation in the panel data.

Hypotheses:

H0: No first-order autocorrelation

H1: First-order autocorrelation exists

Method: Wooldridge test for autocorrelation in panel data (Command: xtserial).

Results: F(1, 15) = 1.958 with Prob > F = 0.1820.

Conclusion: The p-value is greater than 0.05. Therefore, we fail to reject the null hypothesis, indicating no significant evidence of first-order autocorrelation in the model.

3.2.4. Test for Time Fixed Effects

Purpose: To determine whether time-specific events (e.g., economic cycles, policy changes, or global crises) that affect all firms simultaneously should be controlled for in the model.

Hypotheses:

H0: All coefficients for the year dummy variables are equal to zero (No time fixed effects).

H1: At least one year dummy coefficient is different from zero (Time fixed effects exist).

Method: Joint test for all year dummy variables (testparm i.year) after the Fixed Effects regression.

Results: F(13, 96) = 2.74; Prob > F = 0.0024.

Conclusion: Since the p-value (0.0024) is significantly lower than the 0.01 threshold, we reject the null hypothesis at the 1% significance level. H0 is rejected at the 1% level. Time-fixed effects are significant, necessitating the inclusion of year dummies to prevent omitted variable bias.

3.2.5. Model Selection and Robustness Remedy

Model Selection:

Purpose: To determine the most appropriate estimation method between Fixed Effects (FE) and Random Effects (RE).

Method: Hausman Specification Test.

Results: Prob > chi^2 = 0.0000.

Conclusion: The null hypothesis is rejected at the 1% significance level, indicating that the Fixed Effects (FE) model is more appropriate than the Random Effects model.

Robustness Remedy:

To address the confirmed heteroskedasticity (found in 3.2.2) and the unbalanced nature of the small sample (N=21), the study employs **Huber-**

White robust standard errors clustered at the firm level for all subsequent regressions.

3.3. Regression Results for the Full Sample

3.3.1. Goodness-of-Fit and Statistical Significance Evaluation

Goodness-of-Fit Test:

The Fixed Effects model for the full sample yielded an R-squared (within) of 0.5366 for the ROA model and 0.4821 for the ROE model. This suggests that the independent variables explain approximately 53.7% and 48.2% of the variation in profitability within firms over time, respectively. The F-test for both models is significant at the 1% level, confirming the overall validity of the regression.

Statistical Significance (P-value Test):

To determine the statistical significance of each individual regression coefficient, this study relies on the standard P-value approach. A variable is considered to have a statistically significant impact on the dependent variable if its P-value is less than the chosen significance levels (α). In the regression tables, the significance levels are denoted by asterisks:

*** $p < 0.01$ (Significant at the 1% level - Strongest evidence)

** $p < 0.05$ (Significant at the 5% level - Strong evidence)

*$p < 0.10$ (Significant at the 10% level - Weak evidence)

If the P-value exceeds 0.10, the null hypothesis is not rejected, meaning the variable has no significant impact.

Table 3. Impact of corporate governance and financial controls on pre-tax ROE and ROA

Variables	ROA	ROE
boardsize	-0.0181*	-0.0239
independentboardmembe	0.0050	-0.0176

r		
femaleboardmember	0.0298*	0.0186
duality	-0.0344	-0.0293
numberofboardmeeting	0.0019***	0.0027**
chair_gender	0.0447	0.0388
chair_age	0.0006	-0.0007
ceo_gender	0.0582**	0.0714
ceo_age	0.0057***	0.0116***
ln_size	0.0315**	0.0883**
leverage	-0.2727***	-0.3277***
sales_growth	0.0076*	0.0141
foreign_ownership	-0.0414	-0.0402
big4	0.0602**	0.1247**
fixed_asset_ratio	-0.0709	0.0022

Source: Aggregated results from Stata

3.3.2. Impact of CG on Pre-tax ROA

Corporate Governance Variables:

Number of board meetings: Shows a highly significant positive impact (ha= 0.0019, $p < 0.01$).

CEO age: Exhibits a highly significant positive correlation (ha= 0.0057, $p < 0.01$).

CEO gender (1=Male): Has a significant positive impact (ha= 0.0582, $p < 0.05$), indicating higher ROA for firms with male CEOs in this sample.

Board size: Shows a negative impact at the 10% significance level (ha= -0.0181, $p < 0.10$).

Female board member: Shows a positive impact at the 10% significance level (ha= 0.0298, $p < 0.10$).

Statistically insignificant variables ($p > 0.10$): independentboardmember, duality, chair_gender, and chair_age.

Control Variables:

Leverage: Exerts the strongest negative impact on ROA (ha= -0.2727, $p < 0.01$).

Big4 auditing: Shows a significant positive impact (ha= 0.0602, $p < 0.05$).

Firm size (ln_size): Shows a significant positive correlation (ha= 0.0315, $p < 0.05$).

Sales growth: Positively significant at the 10% level (ha= 0.0076, $p < 0.10$).

Fixed asset ratio: Does not show statistical significance ($p > 0.10$).

3.3.3. Impact of CG on Pre-tax ROE

Corporate Governance Variables:

CEO age: Maintains a highly significant positive impact (ha= 0.0116, $p < 0.01$).

Number of board meetings: Remains positively significant at the 5% level (ha= 0.0027, $p < 0.05$).

Statistically insignificant variables ($p > 0.10$): boardsize, femaleboardmember, independentboardmember, duality, chair_gender, chair_age, and ceo_gender.

Control Variables:

Leverage: Exhibits a powerful negative impact (ha= -0.3277, $p < 0.01$).

Big4 auditing: Shows a significant positive impact (ha= 0.1247, $p < 0.05$).

Firm size (ln_size): Positively significant at the 5% level (ha= 0.0883, $p < 0.05$).

Sales growth: Shows a stronger positive significance compared to the ROA model (ha= 0.0141, $p < 0.05$).

Foreign ownership and Fixed asset ratio: No statistical significance found ($p > 0.10$).

3.4. Comparative Analysis Results

3.4.1. Performance Comparison: ROA vs ROE

Variable Significance: More corporate governance variables are statistically significant in the ROA model (5 variables) compared to the ROE model (2 variables). Variables such as boardsize, femaleboardmember, and ceo_gender only impact operational efficiency (ROA) but not equity returns (ROE).

Directional Consistency: Factors such as CEO age, Number of board meetings, Leverage, Firm size, and Big4 auditing show consistent directional impacts across both profitability metrics.

Magnitude of Impact: The absolute values of the coefficients are generally larger in the ROE model. For instance, the negative coefficient for Leverage is -0.3277 for ROE compared to -0.2727 for ROA, and the positive coefficient for CEO age is 0.0116 for ROE versus 0.0057 for ROA.

3.4.2. Sectoral Comparison: Real Estate vs. Pharmaceutical Industry

Table 4. Impact of corporate governance and financial controls on firms' profitability between pharmaceutical and real estate industry

Variables	Pharmaceutical Firms	Real Estate Firms
boardsize	-0.0251**	-0.0095
independentboardmember	0.0089	0.0146

femaleboardmember	0.0314*	0.0172
duality	-0.1240***	-0.0131
numberofboardmeeting	0.0019	0.0018
chair_gender	0.1108***	-0.0604*
chair_age	-0.0010	0.0017
ceo_gender	0.0864***	-0.0193
ceo_age	0.0074***	0.0030*
ln_size	-0.0035	0.0511
leverage	-0.4315***	-0.1997*
sales_growth	0.0593	0.0077*
foreign_ownership	-0.0805	0.0305
big4	0.1123***	0.0212
fixed_asset_ratio	-0.1812**	-0.1320

Source: Aggregated results from Stata

The sectoral regression analysis reveals distinct differences in how corporate governance and financial controls influence firm performance across the two industries:

Board Size:

In Pharmaceutical firms, board size has a significant negative impact on performance (ha= -0.0251, $p < 0.05$). In Real Estate firms, the effect is also negative but statistically insignificant (ha= -0.0095, $p > 0.10$).

CEO Characteristics (Age and Gender):

CEO Age is a highly significant positive driver for Pharmaceutical firms (ha= 0.0074, $p < 0.01$), whereas it is only marginally significant at the 10% level for Real Estate (ha= 0.0030, $p < 0.10$).

CEO Gender (1=Male) shows a strong positive impact in the Pharmaceutical sector (ha= 0.0864, $p < 0.01$) but is insignificant in the Real Estate sector.

Chair Characteristics:

Chair Gender displays a stark contrast between sectors: it is strongly positive for Pharmaceutical firms (ha= 0.1108, $p < 0.01$) but significantly negative for Real Estate firms at the 10% level (ha= -0.0604, $p < 0.10$).

Duality (CEO is also the Chair): Pharmaceutical firms experience a severe negative impact from duality (ha ha= -0.1240, $p < 0.01$). In contrast, duality does not show a statistically significant effect on the performance of Real Estate firms.

Female Board Member: The presence of female directors is positively significant at the 10% level for Pharmaceutical firms (ha= 0.0314, $p < 0.10$) but is insignificant for Real Estate firms.

Control variables:

Leverage: Leverage is a significant negative factor for both sectors, but the magnitude is more than twice as large in the Pharmaceutical industry (ha= -0.4315, $p < 0.01$) compared to Real Estate (ha= -0.1997, $p < 0.10$).

Big4 Auditing: Hiring a Big4 auditor yields a highly significant positive result for Pharmaceutical firms (ha= 0.1123, $p < 0.01$), while it has no significant impact on the profitability of Real Estate firms.

Sales Growth:Sales growth is positively significant for Real Estate firms at the 10% level (ha= 0.0077, $p < 0.10$) but is statistically insignificant for the Pharmaceutical sector.

Fixed Asset Ratio: A higher ratio of fixed assets significantly reduces profitability in Pharmaceutical firms (ha= -0.1812, $p < 0.05$), while the impact is negative but insignificant in the Real Estate sector.

In summary, the Pharmaceutical industry appears much more sensitive to corporate governance mechanisms (such as Duality, Board Size, and CEO/Chair characteristics) and audit quality (Big4). On the other hand, Real Estate firms in this sample show fewer significant governance drivers, with their performance being primarily influenced by financial factors like Leverage and Sales Growth.

3.5. Hypothesis Testing Summary

Table 5. Hypothesis summary

Hypothesis	**Expected Sign**	**ROA Result (Full Sample)**	**ROE Result (Full Sample)**	**Sectoral Difference (Pharmaceutical vs. Real Estate)**	**Status**
H1: Board Size	Negative (-)	-0.0181*	-0.0239	Significant in Pharmaceutical (**), Insignificant in Real Estate	Partially Supported
H2: Board Independence	Positive (+)	0.0050	-0.0176	Insignificant in both sectors	Rejected
H3:	Positive	0.0019*	0.0027*	Positive but	Supported

Frequency of Board Meetings	(+)	**	*	insignificant in small sub-samples	
H4: Gender Diversity (Female Members)	Positive (+)	0.0298*	0.0186	Significant in Pharmaceutical (*), Insignificant in Real Estate	Partially Supported
H5: CEO Duality	Negative (-)	-0.0344	-0.0293	Strongly significant negative impact in Pharmaceutical (***)	Supported (Pharmaceutical only)
H6: CEO Age	Positive (+)	0.0057***	0.0116***	Highly significant in Pharmaceutical (**), Weakly significant in Real Estate ()	Supported
H7: Big 4 Auditors and Foreign Ownership	Positive (+)	0.0602**	0.1247**	Big 4 impact is highly significant in Pharmaceutical (***)	Supported (Big 4)
H8: Sectoral	Difference	(N/A)	(N/A)	Significant differences in	Supported

Differences				coefficients and significance levels	

**Significant at 10%; **Significant at 5%; ***Significant at 1%*

Source: Aggregated results from Stata and group consensus

CHAPTER 4. DISCUSSION

4.1. Interpretation of Results

4.1.1. Discussion on Corporate Governance and Firm Profitability

The empirical results suggest that several corporate governance mechanisms are associated with firm profitability (Peng et al., 2024). Among internal governance variables, board diligence plays an important role. The number of board meetings is positively associated with profitability in the full sample (ROA: $\beta = 0.0019$, ***$p < 0.01$; ROE: $\beta = 0.0027$), suggesting that more frequent board engagement may strengthen monitoring and improve managerial discipline (Vafeas, 1999; Klapper and Love, 2004). From an agency theory perspective, active boards are better able to supervise management decisions and ensure more efficient allocation of resources (Jensen and Meckling, 1976; Ndubuisi, 2023).

Executive characteristics also appear relevant. CEO age shows a strong positive relationship with profitability (ROA: $\beta = 0.0057$, ***$p < 0.01$; ROE: $\beta = 0.0116$, ***$p < 0.01$). This finding indicates that more experienced executives may possess greater managerial capability and strategic stability, which can contribute to stronger firm performance (Hambrick and Mason, 1984). This interpretation aligns with the upper echelons theory, which argues that managerial characteristics influence strategic outcomes (Hambrick and Mason, 1984).

By contrast, structural board variables show weaker effects. Board size is negatively associated with ROA at the 10% level ($\beta = -0.0181$), supporting the argument that larger boards may suffer from coordination and decision-making inefficiencies (Jensen, 1993). Female board representation is positively related to ROA ($\beta = 0.0298$, *$p < 0.10$), suggesting that gender diversity may enhance board deliberation and decision quality, although the evidence remains modest (Campbell and Mínguez-Vera, 2008).

External governance mechanisms also matter. Big4 auditing is positively associated with profitability (ROA: $\beta = 0.0602$, $^{**}p < 0.05$; ROE: $\beta = 0.1247$, $^{**}p < 0.05$), indicating that higher audit quality may enhance financial transparency and reduce information asymmetry.

Finally, control variables highlight the importance of financial structure. Leverage is strongly and negatively related to both ROA and ROE (ROA: $\beta = -0.2727$; ROE: $\beta = -0.3277$, $^{***}p < 0.01$), reflecting interest burdens and financial risk (Claessens and Yurtoglu, 2013). In contrast, firm size is positively associated with profitability (ROA: $\beta = 0.0315$; ROE: $\beta = 0.0883$), suggesting potential scale advantages (Yulianingsih and Kurniawan, 2024).

4.1.2. Discussion on the Differences between Profitability Proxies (ROA vs. ROE)

Empirical results show corporate governance (CG) mechanisms consistently influence firm performance, but effects vary in magnitude and significance between Pre-tax ROA and Pre-tax ROE, reflecting their distinct financial dimensions.

Operational Efficiency vs. Financial Leverage

Internal governance variables like boardsize ($\beta = -0.0181$) and duality ($\beta = -0.0344$) retain negative signs across models. ROA better captures managerial efficiency in asset utilization, mitigating the agency theory "quiet life" hypothesis. ROE, however, amplifies leverage effects, with high debt ($\beta = -0.2727$ on ROA) eroding margins via interest burdens and volatile "leverage effect" mediation during downturns.

Monitoring Quality and Information Asymmetry

Big4 auditing ($\beta = 0.0602$) positively and significantly impacts ROA, signaling financial integrity per signaling theory by enhancing accrual quality and reducing asymmetry. This effect is more robust for ROA than ROE, which remains susceptible to equity "window-dressing."

Alignment of Interests and Risk-Taking

Board independence and executive traits (e.g., ceo_age, gender diversity) show more consistent positive effects on ROE, aligning managerial and shareholder interests. Diverse or older boards promote balanced risk-taking, safeguarding long-term value without excessive ROE pursuit.

4.1.3. Sectoral Discussion: Why Real Estate and Pharma Differ?

The sectoral fixed-effects regressions provide clear evidence that the governance-profitability relationship is industry-contingent.

Profitability is more responsive to internal arrangements. CEO duality shows a strong negative association with ROA ($\beta = -0.1240$, $p < 0.01$), suggesting power concentration weakens board monitoring and operational efficiency. Board size is also significantly negative ($\beta = -0.0251$, $p < 0.05$), consistent with coordination and decision-making costs in large groups. Leadership characteristics and Big4 auditing remain statistically significant, indicating transparent governance and credible assurance drive performance in this compliance-intensive sector.

Firms exhibit fewer significant internal governance drivers, with profitability linked to financial and cycle factors. Leverage is negative and weakly significant ($\beta = -0.1997$, $p < 0.10$), while sales growth is positive ($\beta = 0.0077$, $p < 0.10$), reflecting the capital-intensive, project-based nature. Weaker governance coefficients likely stem from macro-financial influences (credit availability, interest rates, property cycles) dominating within-firm variation.

Overall, these findings support the "no one-size-fits-all" implication: the effectiveness of governance mechanisms depends on sector characteristics. In pharmaceuticals - where regulatory compliance, quality control, and long-term strategic discipline are central - clear separation of roles, effective board structure, and high audit quality are more strongly associated with profitability. In real estate - where financing structure and market cycles play a larger role -

financial discipline and growth dynamics may overshadow internal governance design in explaining profitability.

4.2. Comparison with Prior Studies

The study's empirical results offer a comprehensive understanding of the relationship between governance and profitability, both supporting and deviating significantly from previous research, especially in light of the distinctive institutional context of Vietnam's HNX.

4.2.1. Internal Governance and the Efficiency of Monitoring

Board meeting frequency has a favorable and significant impact on both ROA and ROE, according to the data, which is consistent with Vafeas's (1999) work and the Agency Theory. This implies that active board participation serves as a vital replacement for weaker external legal safeguards for Vietnamese listed companies, guaranteeing that managerial decisions accurately reflect the interests of shareholders. This outcome supports earlier research in emerging economies that suggests formal board structure is frequently less important than board diligence.

However, there is a noticeable difference in terms of board size. Our findings for the pharmaceutical industry indicate a substantial negative impact ($\beta = -0.0251$, $p < 0.05$), despite Resource Dependence Theory (Pfeffer & Salancik, 1978) suggesting that larger boards bring greater resources. This confirms "coordination malfunction" theory (Jensen, 1993), which holds that huge boards can become bureaucratic and impede the agility required for innovation-driven profitability in highly regulated and complex businesses like pharmaceuticals. The insignificance of board size in the real estate sector, on the other hand, indicates that internal board composition is not as important to firm performance in this sector as capital cycles and land bank advantages.

4.2.2. Executive Characteristics and the Role of Human Capital

The confirmation of the role of CEO age (proxied by birth year) is a significant contribution of this study. The Upper Echelons Theory is supported by the

positive relationship between profitability and maturity (Hambrick & Mason, 1984). This is consistent with earlier Vietnamese research' findings, which indicate that older CEOs make use of their established "social capital" and professional networks-factors that are especially important for project approvals and negotiating Vietnam's complicated regulatory settings.

Additionally, the Agency Theory's caution against power concentration is strongly supported by the negative effect of CEO Duality in the Pharmaceutical sample ($\beta = -0.124$, $p < 0.01$). This outcome is in line with Fama and Jensen's (1983) assertion that organizations with strong information asymmetry, like those in the science-based pharmaceutical industry, must keep decision management and control separate.

4.2.3. Gender Diversity and External Signaling Mechanisms

The diversity-performance relationship suggested by Campbell and Mínguez-Vera (2008) is empirically supported by the positive significance of female board members on ROA ($p < 0.1$). Our findings imply that gender diversity in the boardroom enhances the quality of strategic monitoring and decision-making, in contrast to several earlier Vietnamese research that found no significance for gender.

In terms of external governance, the study discovers a strong positive correlation between company performance in the general sample and Big 4 auditing. Hiring a Big 4 auditor in Vietnam acts as a "seal of quality" that reduces information asymmetry, which is consistent with the Signaling Theory (Fan & Wong, 2005). For HNX-listed companies to demonstrate credibility to outside investors and financial institutions, this is especially important.

4.2.4. Sectoral Heterogeneity and Contingency Factors

In the end, the Contingency Theory is strongly supported by the sharp contrast between the pharmaceutical industry, where internal governance is very important, and the real estate industry, where many governance effects are muted. Our results confirm Pfeffer and Salancik's (1978) reasoning that the

efficacy of corporate governance depends on the industry. Real estate companies are more susceptible to macroeconomic factors, such as the significant negative impact of financial leverage seen in our models ($\beta = -0.316$), making internal governance a secondary factor in their immediate profitability outcomes compared to financial structure, whereas pharmaceutical companies heavily rely on internal monitoring to safeguard R&D investments.

CHAPTER 5. RECOMMENDATIONS

5.1. Implications for Corporate Strategists and Board Restructuring

The findings indicate that corporate governance mechanisms are more strongly associated with operational profitability (Pretax ROA) than with equity returns (Pretax ROE), while financial leverage has the strongest negative effect. These results suggest several managerial implications. Board structure should be optimized rather than expanded, prioritizing director competence, independence, and industry expertise. Active board oversight can improve asset utilization, although governance effectiveness depends on meeting quality rather than frequency alone. Leadership structure also matters, as CEO experience enhances profitability while CEO duality may weaken monitoring, supporting separation of the CEO and Chair roles. Finally, prudent debt management is essential, given the significant negative impact of leverage on profitability.

5.2. Recommendations for Policymakers and Regulators

The findings provide important policy implications, as governance mechanisms are linked to operational performance. Regulators should strengthen governance standards by improving disclosure on board structure, leadership roles, meeting frequency, and audit practices to enhance market transparency. The positive relationship between Big 4 auditing and profitability highlights the importance of audit quality for market credibility and investor confidence. The results also suggest industry-specific differences, indicating that governance regulations should consider sectoral characteristics rather than applying uniform requirements across industries.

5.3. Recommendations for Investors

For investors, corporate governance indicators are important when evaluating firms, especially regarding operational efficiency. Factors such as board diligence, leadership structure, audit quality, and capital structure discipline may signal long-term sustainability. However, ROE is highly leverage-

sensitive, as higher debt can boost returns while increasing financial risk. Investors should therefore assess debt sustainability and adopt industry-specific strategies, since governance matters more in pharmaceutical firms, whereas real estate profitability is driven largely by financing cycles and macroeconomic conditions.

CONCLUSION

This study examines the impact of corporate governance on the profitability of 21 HNX-listed firms. Utilizing a Fixed Effects Model with robust standard errors, the empirical evidence confirms that governance mechanisms significantly influence operational efficiency (ROA). Notably, the findings highlight that this relationship is industry-contingent: while internal governance is a primary performance driver in the pharmaceutical sector, real estate firms are more susceptible to macroeconomic cycles and capital structure dynamics.

However, the study is not without limitations. The relatively small sample size may limit the generalizability of the findings to the entire Vietnamese market. Furthermore, the reliance on accounting-based proxies might not capture the full spectrum of market-based performance or tax-shield complexities.

Future research should expand the dataset to include HOSE-listed firms and integrate qualitative governance dimensions, such as board expertise and diversity beyond gender. Investigating the long-term impact of governance "quality" rather than mere "structure" would provide more profound insights into corporate sustainability in emerging markets.

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APPENDIX

APPENDIX 1: PAIRWISE CORRELATIONS

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)
(1) pretax_roa	1																
(2) pretax_roe	0.8267***	1															
(3) boardsize	0.2628***	0.2250***	1														
(4) independent board member	-0.1673**	-0.1374**	0.0213	1													
(5) female board member	0.0590	-0.0118	0.2406***	0.1060	1												
(6) duality	0.0490	0.0506	-0.0692	-0.1457**	0.0909	1											
(7) number of board meeting	-0.0487	-0.0203	-0.1082	-0.0967	-0.0596	-0.1514**	1										
(8) chair_gender	0.0577	0.0641	-0.1111*	-0.1419**	-0.302***	-0.0230	0.0610	1									
(9) chair_age	0.1895***	0.1678***	0.2525***	-0.1392**	0.1807***	-0.0135	-0.0061	-0.1096*	1								
(10) ceo_gender	-0.0308	-0.0016	0.0362	-0.1236*	-0.296***	0.0150	-0.1084	0.1090*	-0.1277*	1							
(11) ceo_age	0.3988***	0.3354***	0.2580***	-0.1785**	0.1384**	0.1663**	-0.0812	-0.0417	0.5588***	-0.197***	1						
(12) ln_size	-0.1814***	-0.1639**	-0.1297**	0.5928***	0.1937**	-0.0805	-0.1205*	-0.337***	-0.1059*	0.0216	-0.1333**	1					
(13) leverage	-0.1501**	0.1869***	0.1487**	0.1316*	-0.0233	0.0094	-0.0114	-0.1185*	0.0620	-0.0285	0.0119	0.1487**	1				
(14) sales growth	0.0263	0.0551	0.0074	-0.0565	-0.0327	0.1638**	-0.0033	-0.0522	-0.0677	-0.0781	-0.0999	-0.1034	-0.0036	1			
(15) foreign ownership	-0.1422**	-0.1939***	-0.2340***	0.1149*	-0.1397**	-0.1198*	-0.0601	-0.0692	-0.0864	0.2882***	-0.1350**	0.1748***	-0.1811***	0.0540	1		
(16) big4 auditing	-0.177***	-0.185***	-0.1342**	0.3306***	-0.0708	-0.1152*	-0.1146	-0.0141	-0.0701	-0.1541**	-0.307***	0.2426***	-0.0418	0.0654	0.0420	1	
(17) PPE	0.0901	0.0096	-0.0647	0.2956***	0.0875	0.1866***	-0.289***	-0.182***	0.0955	0.0854	0.0883	0.5476***	0.0601	-0.1480**	0.0585	0.1923***	1

****p<0.01, **p<0.05, *p<0.1*

APPENDIX 2: LIST OF ABBREVIATIONS

Abbreviation	Meaning
CG	Corporate Governance
ROA	Return on Assets
ROE	Return on Equity
HNX	Hanoi Stock Exchange
OLS	Ordinary Least Squares
POLS	Pooled Ordinary Least Squares
FE (FEM)	Fixed Effects (Fixed Effects Model)
RE (REM)	Random Effects (Random Effects Model)
SE	Standard Error
CLSE	Clustered Standard Errors
VIF	Variance Inflation Factor

EBT	Earnings Before Tax
CEO	Chief Executive Officer
CEO Duality	CEO also serves as Board Chair
Big4	Big Four Auditing Firms
FO	Foreign Ownership
Lev	Leverage
PPE	Property, Plant and Equipment
R&D	Research and Development
Ln	Natural Logarithm
Hausman	Hausman Specification Test (FE vs. RE)
Wald (xttest3)	Modified Wald Test for Groupwise Heteroskedasticity (FE)
Wooldridge (xtserial)	Wooldridge Test for Autocorrelation in Panel Data

APPENDIX 3: LIST OF VARIABLES

Variable	Description	Unit	Variable type
pretax_roa	Pre-tax ROA = Earnings before tax / Average total assets	Ratio (decimal)	Dependent
pretax_roe	Pre-tax ROE = Earnings before tax / Average	Ratio (decimal)	Dependent

	shareholders' equity		
boardsize	Total number of board members	Count (persons)	Corporate governance
independentboardmember	Number of independent directors	Count (persons)	Corporate governance
femaleboardmember	Number of female board members	Count (persons)	Corporate governance
duality	CEO duality dummy (1 if CEO is also Chair)	Dummy (0/1)	Corporate governance
numberofboardmeeting	Number of board meetings per year	Count (meetings/ year)	Corporate governance
chair_gender	Chair gender (male=1, female=0)	Dummy (0/1)	Corporate governance
chair_age	Chair age (year − chair_birthyear)	Years	Corporate governance
ceo_gender	CEO gender (male=1, female=0)	Dummy (0/1)	Corporate governance
ceo_age	CEO age (year − ceo_birthyear)	Years	Corporate governance
ln_size	Firm size = ln(total assets)	Log value	Control
leverage	Total liabilities / Total assets	Ratio (decimal)	Control
sales_growth	(Revenue_t − Revenue_{t−1}) /	Ratio (decimal)	Control

	Revenue_{t−1}		
foreign_ownership	Foreign ownership dummy (1 if foreign ownership present)	Dummy (0/1)	Control (external governance proxy)
big4	Big4 auditor dummy (1 if audited by Big4)	Dummy (0/1)	Control (external governance proxy)
fixed_asset_ratio	Gross PPE / Total assets	Ratio (decimal)	Control
BDS	Industry dummy (1=Real Estate, 0=Pharmaceutical)	Dummy (0/1)	Sector indicator

Note: pretax_roa, pretax_roe, leverage, sales_growth, and fixed_asset_ratio are winsorized at the 1st and 99th percentiles; ln_size is log-transformed.

APPENDIX 4: MULTICOLLINEARITY TEST (VIF)

Variable	VIF	1/VIF
boardsize	2.33	0.429117
independentboardmember	3.76	0.266175
femaleboardmember	2.39	0.418761
duality	2.00	0.499595
numberofboardmeeting	1.50	0.666802
chair_gender	1.92	0.522048
chair_age	2.15	0.464737
ceo_gender	1.98	0.504786

ceo_age	2.01	0.496594
ln_size	4.08	0.244884
leverage	1.46	0.684430
sales_growth	1.19	0.840258
foreign_ownership	1.64	0.610789
big4	1.60	0.625167
fixed_asset_ratio	3.69	0.271035
year 2012	4.24	0.235733
year 2013	4.27	0.234403
year 2014	5.03	0.198985
year 2015	6.35	0.157540
year 2016	6.99	0.143097
year 2017	6.73	0.148501
year 2018	7.09	0.141037
year 2019	8.72	0.114722
year 2020	8.94	0.111917
year 2021	10.67	0.093748
year 2022	10.64	0.094019
year 2023	9.26	0.107977
year 2024	10.99	0.091028

APPENDIX 5: DIAGNOSTIC TESTS

Test	Statistic	Result
Hausman specification test (FE vs. RE)	chi2(17) = 85.80	Prob > chi2 = 0.0000
Modified Wald test for groupwise	chi2(18) = 1762.34	Prob > chi2 = 0.0000

heteroskedasticity		
Wooldridge test for autocorrelation	F(1,15) = 1.958	Prob > F = 0.1820